

Cloud Computing (CS-4075)

Course Instructor(s)

Dr. Basharat Hussain

Section(s)

8A, 8B - (CS)

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 55

Total Questions: 3

Date: (Monday) Feb 23, 2026

Roll No

Course Section

Student Signature

YOU MUST READ THESE INSTRUCTIONS.

Instructions:

1. DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.
2. Attempt ALL questions on the provided **answer sheet** except MCQs (if any). Read the question carefully, understand the question, and then attempt it.
3. Return the **question paper** along with the answer sheet .
4. Attempt all the questions.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.

[This is the CLO 2: Apply cloud computing knowledge in various real world use cases (situations).]

[This is the CLO 3: Analyze and architect a cloud-based solution in a real-world use case.]

Question # 1

[Avg-attempt-minutes= 20, Marks = 15]

Topic: Cloud Infrastructure Design

Part A: Network Design (10 Marks)

A retail startup is moving their website to AWS. They require a secure, two-tier architecture that can handle traffic spikes and maintains a private database. You must design the network to be highly available across two Availability Zones (AZs). You are required to design a Virtual Private Cloud (VPC) architecture in Amazon Web Services (AWS) for a three-tier web application.

- VPC CIDR Block: 172.10.0.0/? (must accommodate all required subnets)
- Public Subnets: Each must support at least 1020 usable IP addresses.
- Private Subnets: Each must support at least 50 machines.
- Two Availability Zones (AZ-A and AZ-B) must be used.

You must:

THIS ANSWER SHEET IS THE ANSWER SHEET FOR THE NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES ISLAMABAD CAMPUS

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Design the inbound and outbound rules for both security groups under the following requirements:

1. Application servers must allow HTTP (TCP 80) and HTTPS (TCP 443) access from all IPv4 addresses.
2. Linux-based instances must allow SSH (TCP 22) access only from the organization's public IPv4 network range.
3. Database servers must allow Microsoft SQL Server traffic (TCP 1433) only from application servers.
4. All necessary return traffic must be permitted.

Hint: Clearly define:

- Inbound rules for SG1 and SG2
- Outbound rules for SG1 and SG2
- Columns: Source, Protocol, Port Range, Brief description for each rule

Part B: Network ACL Configuration (10 Marks)

Based on your VPC and Security Group design, configure Network Access Control Lists (NACLs) for both Public and Private subnets.

- Allow HTTPS from anywhere to Public Subnets.
- Allow SSH (TCP 22) only from 192.0.2.0/24.
- Allow Microsoft SQL Server traffic (TCP 1433) only between Application and Database subnets.
- Allow required ephemeral port ranges for return traffic.
- Explicitly deny all other inbound and outbound traffic.

You must:

- Define inbound and outbound rules with rule numbers.
- Explain how stateless behavior affects your design.
- Clearly distinguish between Security Group and NACL responsibilities.

[This is the CLO 1: Describe the basics of cloud computing and related services.]
[Avg-attempt-minutes= 20, Marks = 20]

Question # 3

- Attempt All the Questions Below on Bubble Sheet at *LAST* Page
- Return question paper with the answer sheet. Don't staple question paper, just put in side the answer sheet.

MCQ 1: In the Shared Responsibility Model, for which aspect of securing the cloud is AWS responsible?

- ☒ a) Security of the cloud
- b) Security for the cloud
- c) Security to the cloud
- d) Security in the cloud

MCQ 2: Which of the following is a correct relationship between regions, Availability Zones, and edge locations?

- a) Data centers contain regions.
- b) Regions contain Availability Zones.
- c) Availability Zones contain edge locations.
- d) Edge locations contain regions.

MCQ 3: Which cloud computing characteristic enables automatic scaling of resources without human intervention?

- a) On-demand self-service
- b) Broad network access
- c) Rapid elasticity
- d) Resource pooling

MCQ 4: An IAM policy contains an explicit Deny for ec2:TerminateInstances, but another attached policy allows it. What is the final result?

- a) Allowed
- ☒ b) Denied
- c) Depends on role hierarchy
- d) Depends on region

MCQ 5: Which condition key ensures that EC2 instances can only be launched in us-east-1?

- a) ec2:Region
- ☒ b) aws:RequestedRegion
- c) aws:RegionKey
- d) ec2:RequestedZone

MCQ 6: An SCP denies s3:DeleteBucket at the organization level. What happens if an IAM user has full S3 access?

- a) User can delete bucket
- b) User can delete only versioned buckets
- ☒ c) Deletion is denied
- d) Deletion allowed in root account

MCQ 7: A VPC with CIDR 10.0.0.0/16 is subnetted into /24 subnets. How many subnets can be created?

- a) 16
- b) 64
- c) 128
- d) 256

MCQ 8: A Security Group allows inbound TCP 80 from 0.0.0.0/0. Outbound rules are default. What is true?

- a) Inbound blocked
- b) Outbound blocked
- ☒ c) Both inbound and outbound allowed
- d) Outbound allowed, inbound restricted

MCQ 9: A NACL denies inbound traffic from 203.0.113.0/24 with rule number 100, but allows all traffic with rule number 200. What happens?

- a) Traffic allowed
- ☒ b) Traffic denied
- c) Depends on port
- d) Evaluated randomly

MCQ 10: If an IAM user does not have MFA enabled and a policy requires `aws:MultiFactorAuthPresent=true`, what happens?

- a) Access allowed
- b) Read-only access
- ☒ c) Access denied
- d) MFA auto-enabled

MCQ 11: A subnet without an Internet Gateway attached to the VPC can:

- ☒ a) Access internet directly
- b) Receive inbound internet traffic
- c) Access internet via NAT Gateway if configured
- ☒ d) Act as public subnet

MCQ 12: How many usable IPs are available in a /27 subnet in AWS?

- a) 32
- ☒ b) 27
- c) 25
- d) 30

MCQ 13: Which policy type can restrict maximum permissions across AWS accounts in an Organization?

- a) IAM Policy
- b) Permissions Boundary
- ☒ c) SCP
- d) Resource Policy

MCQ 14: Security Groups are stateful. This means:

- ☒ a) Inbound rules must match outbound rules
- ☒ b) Return traffic is automatically allowed
- ☒ c) Rules are processed in order
- ☒ d) Explicit deny supported

MCQ 15: A NACL rule table is evaluated in which order?

- a) Random
- b) Descending rule number
- ☒ c) Ascending rule number
- d) By protocol type

MCQ 16: Which Route 53 policy is best for disaster recovery active-passive architecture?

- a) Weighted
- b) Latency-based
- c) Failover
- d) Geoproximity

MCQ 17: CloudFront caches objects at:

- a) Origin server only
- b) Edge locations
- c) Regional VPCs
- d) IAM layer

MCQ 18: If a user is allowed `ec2:*` but denied `ec2:CreateTags`, what can they NOT do?

- a) Launch instances
- b) Stop instances
- ☒ c) Tag instances
- d) Describe instances

MCQ 19: A VPC peering connection does NOT support:

- a) Private IP communication
- ☒ b) Transitive routing
- c) Cross-account connection

d) Cross-region connection

MCQ 20: If CloudFront TTL is set to 0 seconds, what happens?

- a) Content cached permanently
- ☒ b) Content never cached
- c) Content cached at origin
- d) Content encrypted automatically



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Subject 1

Section 1

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Q. 1

Part A

1. Appropriate CIDR block for the VPC : $172.10.0.0/16$

2. CIDR block allocation

• Public Subnet 1 (AZ-A)

∴ We require 1024 Usable IP Addresses

∴ We need 2^{10} Host Address bits⇒ CIDR Block : $172.10.0.0/21$ ✓

• Public Subnet 2 (AZ-B)

CIDR Block : $172.10.8.0/21$ ✓

• Private Subnet 1 (AZ-A)

∴ We require 50 Usable IP Addresses

∴ We need 2^6 Host Address Bits⇒ CIDR Block : $172.10.16.0/26$ ✓

• Private Subnet 2 (AZ-B)

⇒ CIDR Block : $172.10.17.0/26$ ✓

(next page)

3. Assuming the private database needs to access the internet for patch updates since not specified.

- Routing Table for Public Subnet 1 (AZ-A)

Destination	Target
172.10.0.0/16	Local
0.0.0.0/0	igw-id

- Routing Table for Public Subnet 2 (AZ-B)

Destination	Target
172.10.0.0/16	Local
0.0.0.0/0	igw-id

- Routing Table for Private Subnet 1 (AZ-A)

Destination	Target
172.10.0.0/16	Local
0.0.0.0/0	nat-gw-id

- Routing Table for Private Subnet 2 (AZ-B)

Destination	Target
172.10.0.0/16	Local
0.0.0.0/0	nat-gw-id

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Part B

AWS
Cloud

IGW

13/15

Region A7

Availability Zone AZ-A

Public Subnet 1

172.16.0.0/16

NAT-GW-ID

ELB

Private Subnet 1

172.10.16.0/26

Applications

RDS-1

Auto
Scaling
Group

Zone AZ-B

Public Subnet 2

172.10.8.0/26

ELB

Private Subnet 2

172.10.17.0/26

Backup Applications

RDS-2

Q.2

Part A

• SG-1 (Inbound Rules)

Source	Protocol	Type	Port	Effect
0.0.0.0/0	HTTP	TCP	80	Allow
0.0.0.0/0	HTTPS	TCP	443	Allow
Organization's IPv4 range)	SSH	TCP	22	Allow

• SG-1 (Outbound Rules)

IP Range	Protocol	Type	Port	Effect
0.0.0.0/0	*	*	*	Allow

* By default all outbound requests are allowed

• SG-2 (Inbound Rules)

Source	Protocol	Port	Type	Port
sg-1-id			TCP	1443
Organization's IPv4)			SSH	22

• SG-2 (Outbound Rules)

IP Range	Type	Port
sg-1-id	TCP	1443
Organization's IPv4)	SSH	22

Part B

• NACL for Public Subnet (Inbound)

	IP	Protocol	Port	Action
100	0.0.0.0/0	HTTPS	443	Allow
200	192.0.2.0/24	SSH	22	Allow
300	SQL-server-IP	TCP	1433	Allow
*	*	*	*	Deny

18/20

Outbound

	IP	Protocol	Port	Action
100	0.0.0.0/0		443	Allow
200	192.0.2.0/24		22	Allow
300	SQL-server-IP		1433	Allow
*	*		*	Deny

Ethercat.

=> SGT assigned at ~~EG2~~ instance level
 => NACL " " subnet "